

Project Title

Analyzing Smart City Bike Sharing Data using Power BI

Project Objective

Public bike-sharing systems generate continuous data from hundreds of stations across different cities. This project analyzes real-time bike station data to understand station performance, usage efficiency, and operational patterns across cities - applying Power BI skills from data cleaning and modeling to DAX and dashboarding to turn raw data into actionable urban mobility insights. Specifically, the project aims to:

- Identify the highest and lowest performing cities by average station occupancy rate.
- Track total stations, snapshot volume, and network-wide occupancy trend over time (2022–2025).
- Diagnose why specific cities show abnormally high closed-station rates.
- Use historical trend data to anticipate near-term occupancy levels.
- Recommend concrete actions (maintenance, rebalancing, benchmarking) for underperforming cities.

Dataset Description

The dataset consists of continuous, real-time station-level snapshots collected from a public bike-sharing network spanning 24 cities across Europe, covering the period 2022–2025. It combines four tables organized into a star schema:

- Fact_StationStatus - the fact table, containing one row per station snapshot (bike availability, dock capacity, and operational status).
- Dim_Station - station master data, including name, address, coordinates, and banking/bonus flags.
- Dim_City - city and country reference data.
- Dim_Date - calendar/date reference data used for trend and time-based analysis.

Column Description

Table	Column Name	Description
Fact_StationStatus	SnapshotID	Unique identifier for each station status snapshot record
Fact_StationStatus	StationID	Foreign key linking the snapshot to the station in Dim_Station
Fact_StationStatus	DateKey	Foreign key linking the snapshot to the calendar date in Dim_Date
Fact_StationStatus	SnapshotTime	Exact date/time the station status reading was captured
Fact_StationStatus	Hour	Hour of day the snapshot was recorded (0-23)
Fact_StationStatus	BikeStands	Total docking capacity (bike stands) at the station

Fact_StationStatus	AvailableBikeStands	Number of empty docks available to return a bike
Fact_StationStatus	AvailableBikes	Number of bikes currently available for rent at the station
Fact_StationStatus	Status	Operational status of the station at the time of the snapshot (OPEN/CLOSED)
Dim_Station	StationID	Unique identifier for each bike station
Dim_Station	StationName	Name of the bike station
Dim_Station	Address	Street address / location description of the station
Dim_Station	Latitude	Geographic latitude coordinate of the station
Dim_Station	Longitude	Geographic longitude coordinate of the station
Dim_Station	Banking	Flag indicating whether the station has a card payment kiosk
Dim_Station	Bonus	Flag indicating whether the station is a bonus/hilltop station
Dim_Station	CityID	Foreign key linking the station to its city in Dim_City
Dim_City	CityID	Unique identifier for each city
Dim_City	City	Name of the city where stations are located
Dim_City	Country	Country in which the city is located
Dim_Date	DateKey	Unique numeric key identifying each calendar date
Dim_Date	Date	Calendar date
Dim_Date	Year	Calendar year of the date
Dim_Date	Quarter	Calendar quarter (1-4) of the date
Dim_Date	Month	Calendar month number (1-12) of the date
Dim_Date	MonthName	Full name of the month (e.g. January)
Dim_Date	Day	Day of the month
Dim_Date	DayName	Name of the weekday (e.g. Monday)

Data Cleaning & Transformation

The following cleaning and transformation steps were performed in Power Query before loading the data into the model:

- Connected to the raw dataset and imported all four tables (Fact_StationStatus, Dim_Station, Dim_City, Dim_Date) via Transform Data (not a direct Load), to allow verification and correction before loading.

- Corrected column data types across all tables - e.g. SnapshotTime and Date fields were originally read as raw serial numbers and were reset to proper Date/Time and Date types; numeric ID and count fields were set to Whole Number; Latitude/Longitude set to Decimal Number; Banking/Bonus set to True/False.
- Removed duplicate records and trimmed text fields (station names, addresses, city/country names) to eliminate formatting inconsistencies.
- Verified referential integrity of key fields (StationID, CityID, DateKey) used to join the fact table to each dimension table.
- Documented each Applied Step in Power Query Editor and captured screenshots as evidence of the cleaning process (see below).

(Power Query Applied Steps - Dim_Date and Fact_StationStatus)

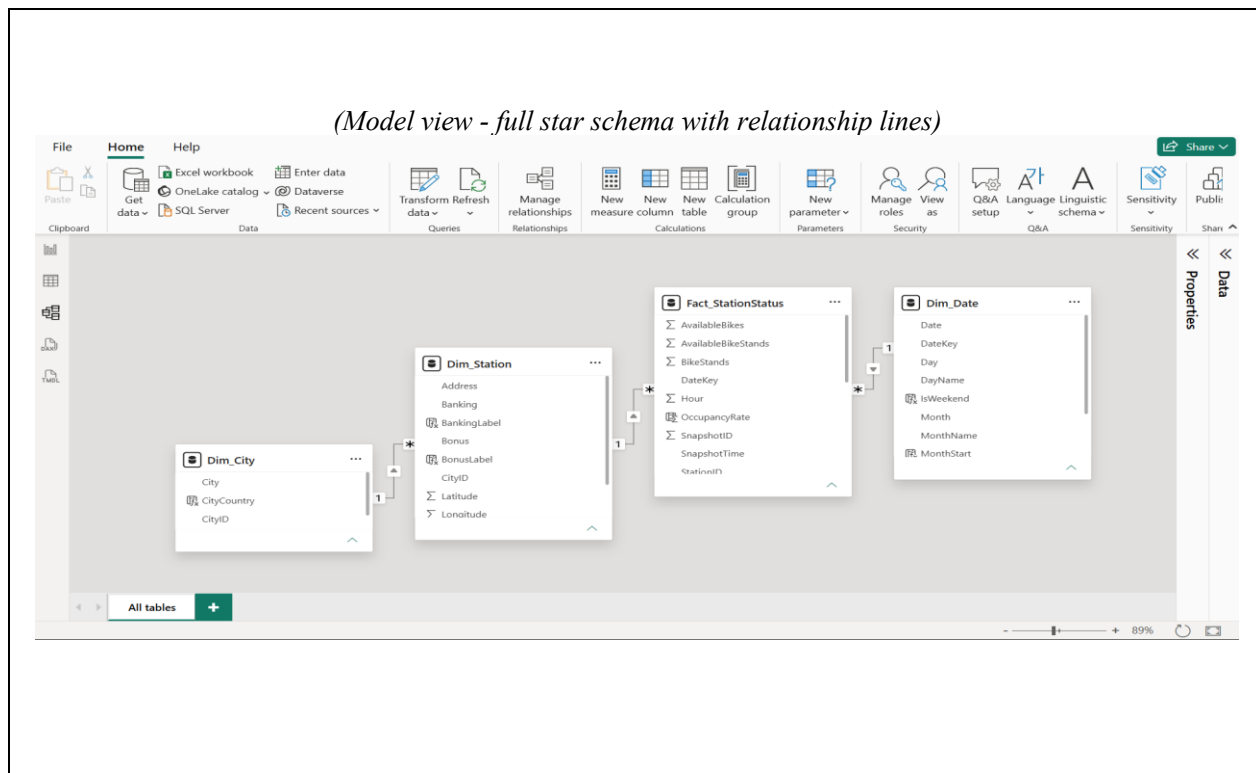
8 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

9 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

Data Modeling - Star Schema

The cleaned tables were assembled into a star schema with Fact_StationStatus at the center and three supporting dimension tables:

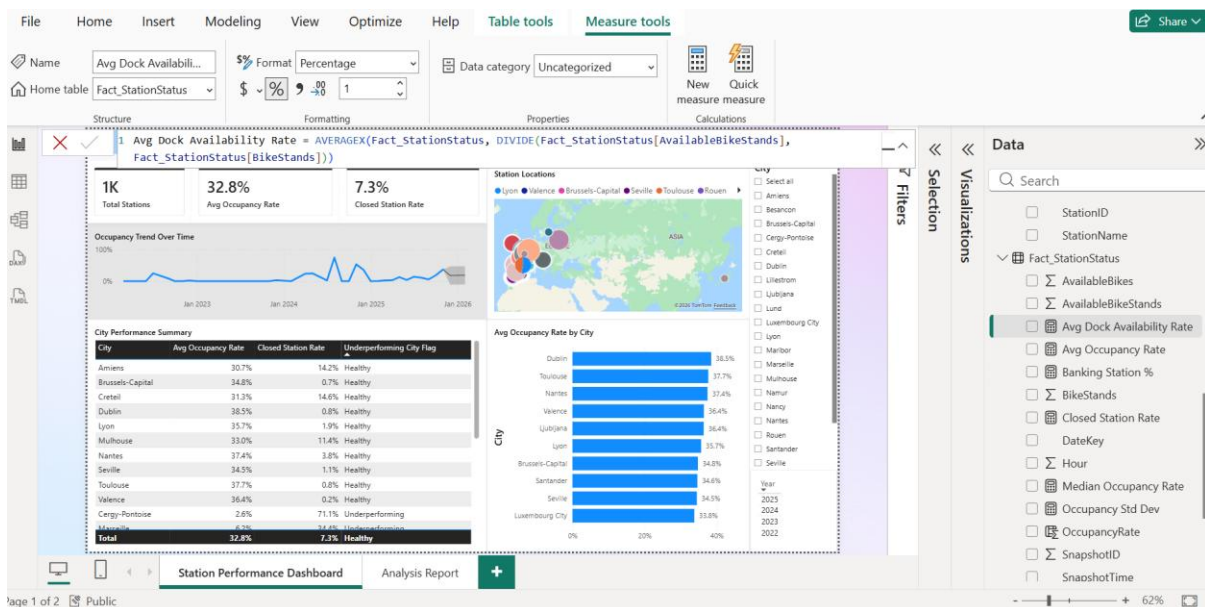
- Dim_Station[CityID] to Dim_City[CityID] - One-to-many, single direction, active
- Fact_StationStatus[StationID] to Dim_Station[StationID] - One-to-many, single direction, active
- Fact_StationStatus[DateKey] to Dim_Date[DateKey] - One-to-many, single direction, active



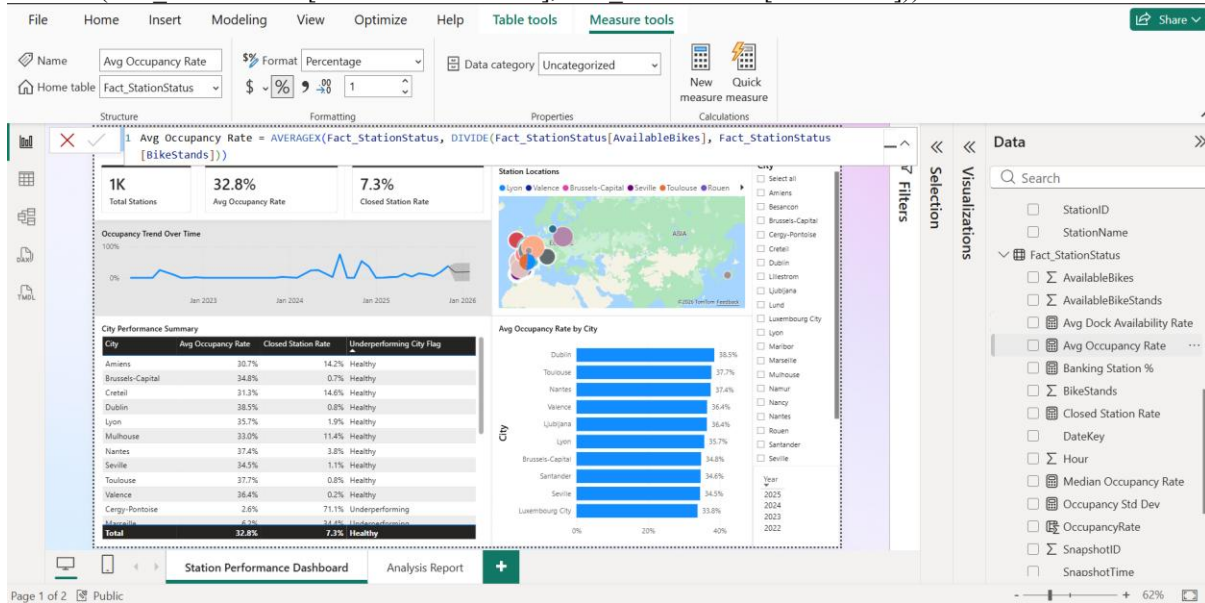
DAX Measures & Calculated Columns

Calculated columns (OccupancyRate, StatusFlag, BankingLabel, BonusLabel, CityCountry, IsWeekend, MonthYearSort/MonthStart) and eleven core measures were built, including Avg Occupancy Rate, Median Occupancy Rate, Occupancy Std Dev, Closed Station Rate, Avg Dock Availability Rate, Total Network Capacity, YoY Occupancy Change, Underperforming City Flag, and Banking Station %.

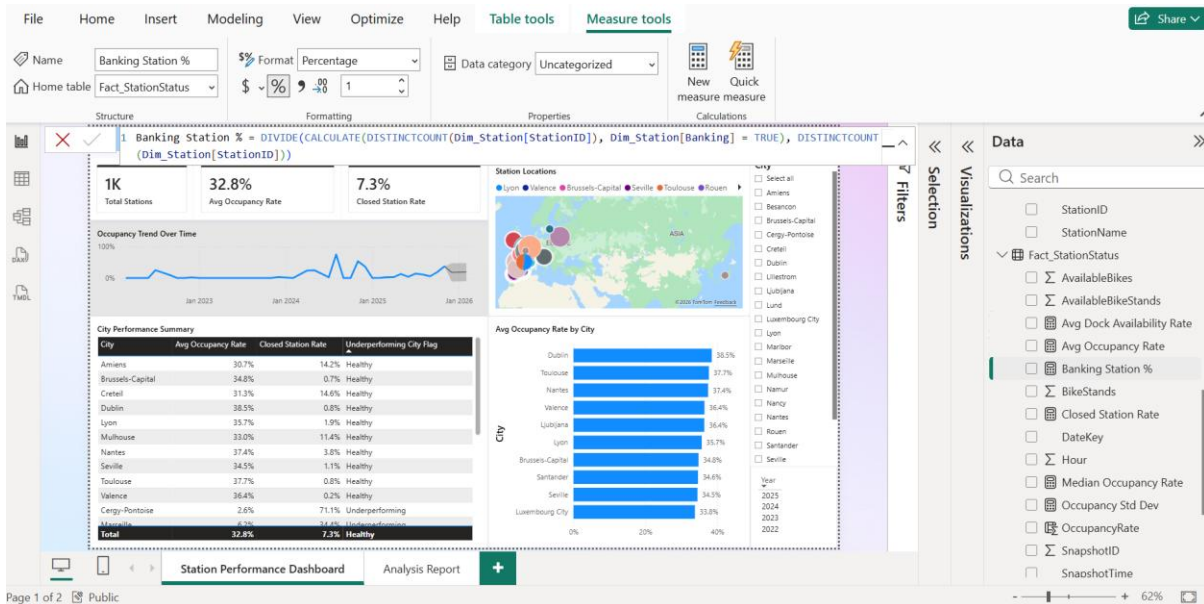
(DAX formula bar)



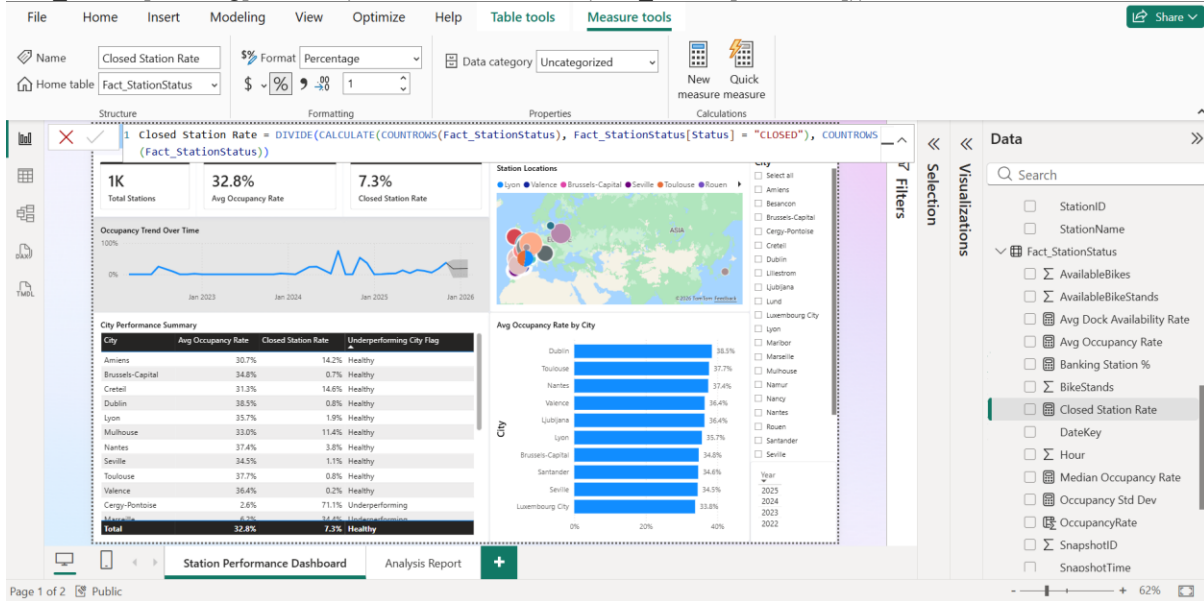
Avg Dock Availability Rate = AVERAGEX(Fact_StationStatus, DIVIDE(Fact_StationStatus[AvailableBikeStands], Fact_StationStatus[BikeStands]))



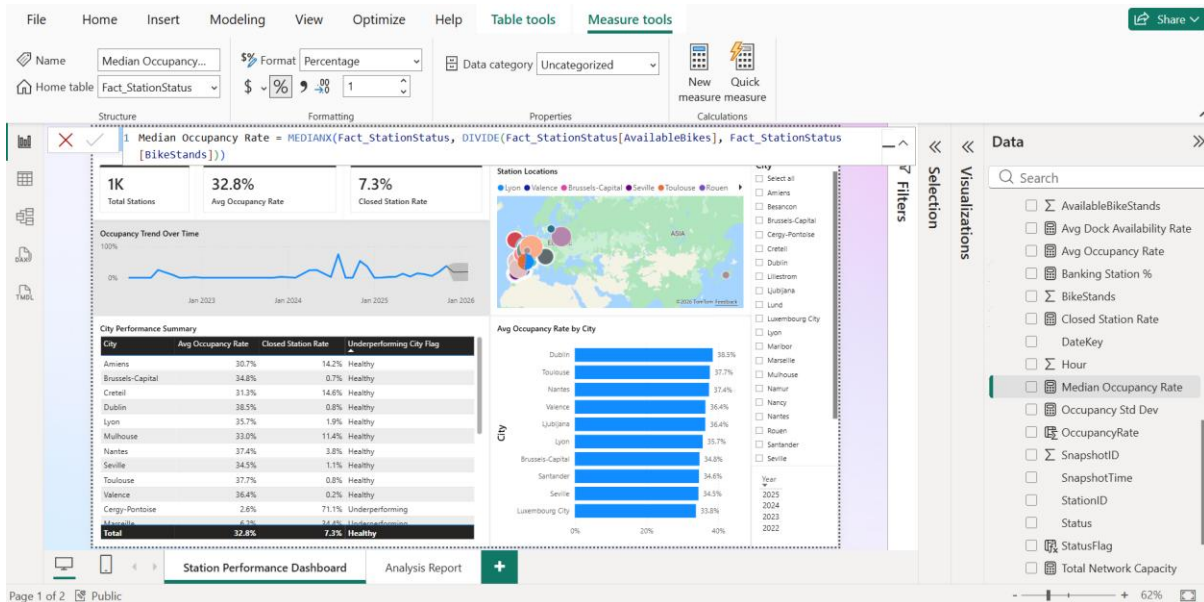
Avg Occupancy Rate = AVERAGEX(Fact_StationStatus, DIVIDE(Fact_StationStatus[AvailableBikes], Fact_StationStatus[BikeStands]))



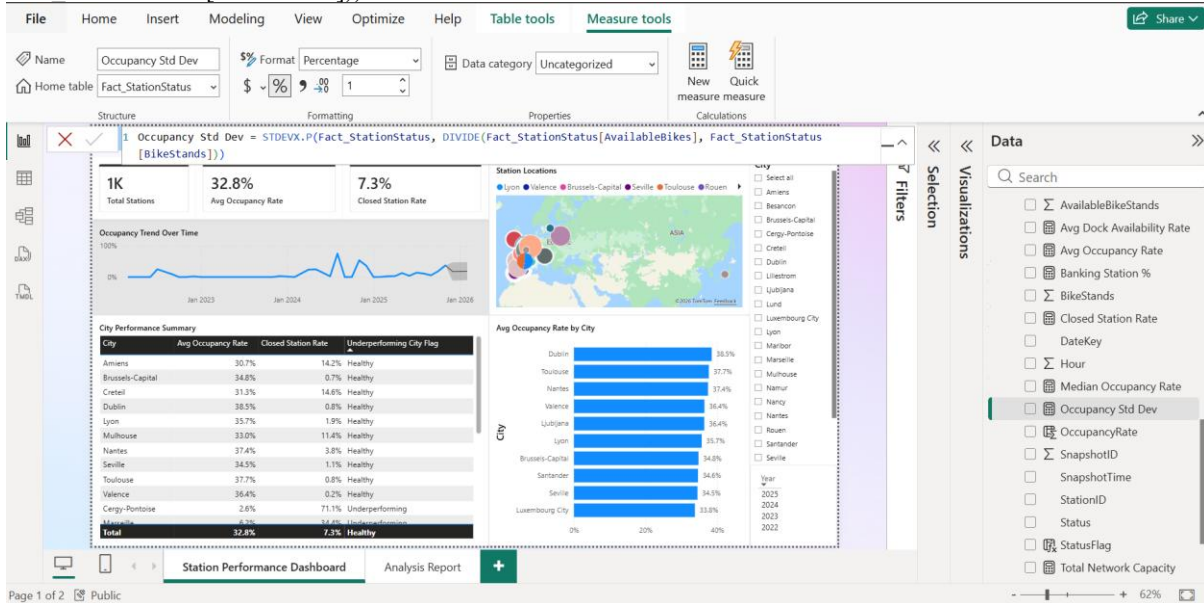
Banking Station % = DIVIDE(CALCULATE(DISTINCTCOUNT(Dim_Station[StationID]), Dim_Station[Banking] = TRUE), DISTINCTCOUNT(Dim_Station[StationID]))



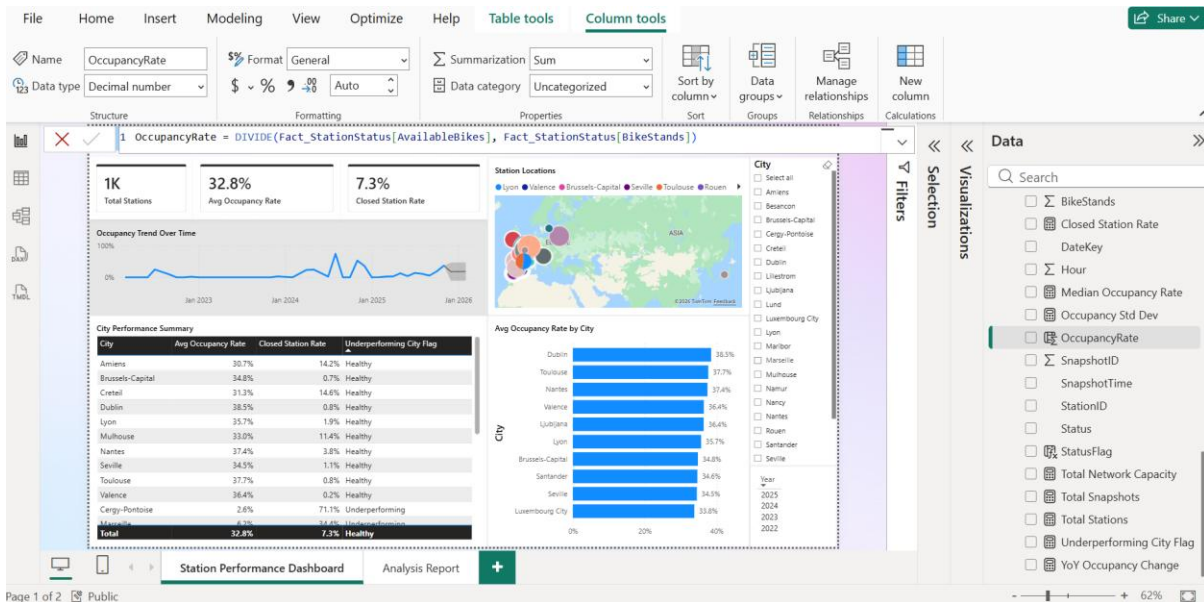
Closed Station Rate = DIVIDE(CALCULATE(COUNTROWS(Fact_StationStatus), Fact_StationStatus[Status] = "CLOSED"), COUNTROWS(Fact_StationStatus))



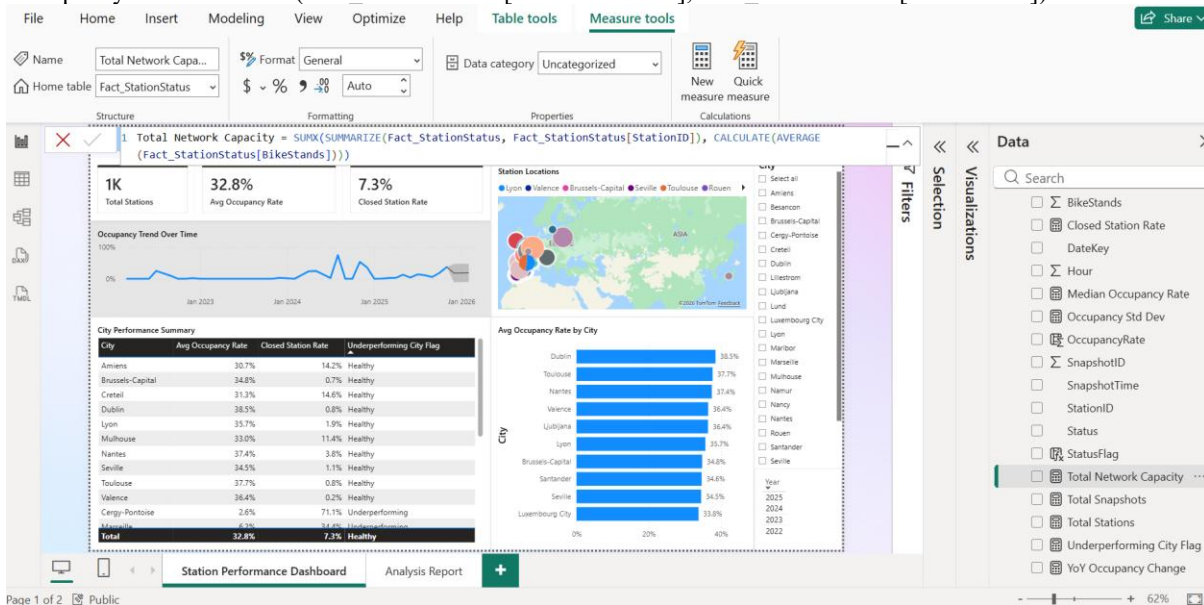
Median Occupancy Rate = MEDIANX(Fact_StationStatus, DIVIDE(Fact_StationStatus[AvailableBikes], Fact_StationStatus[BikeStands]))



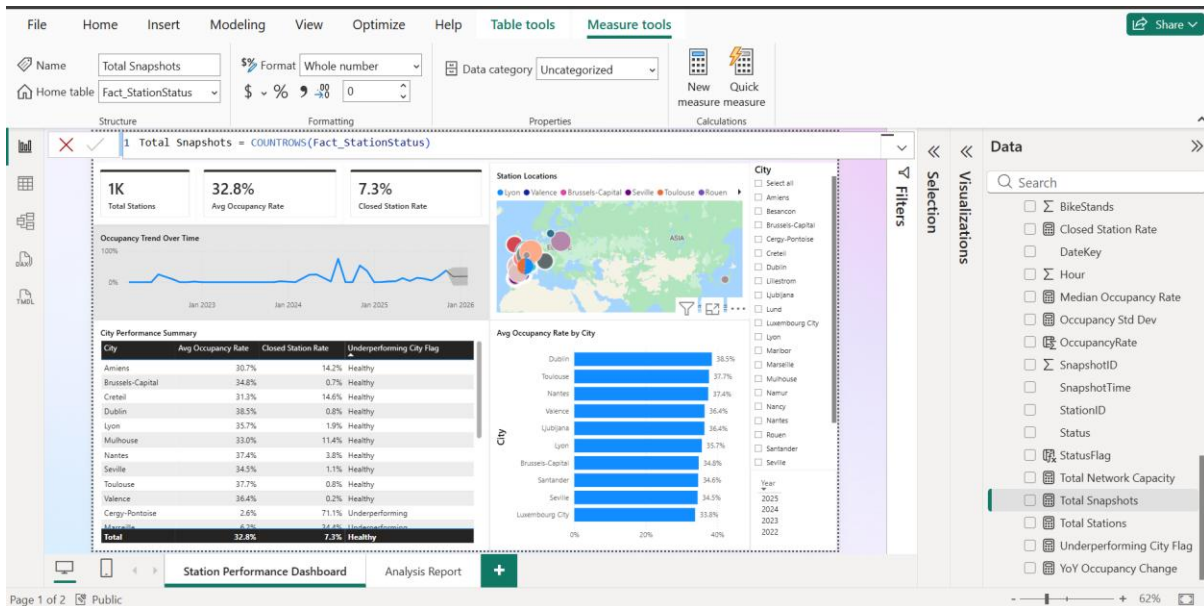
Occupancy Std Dev = STDEVX.P(Fact_StationStatus, DIVIDE(Fact_StationStatus[AvailableBikes], Fact_StationStatus[BikeStands]))



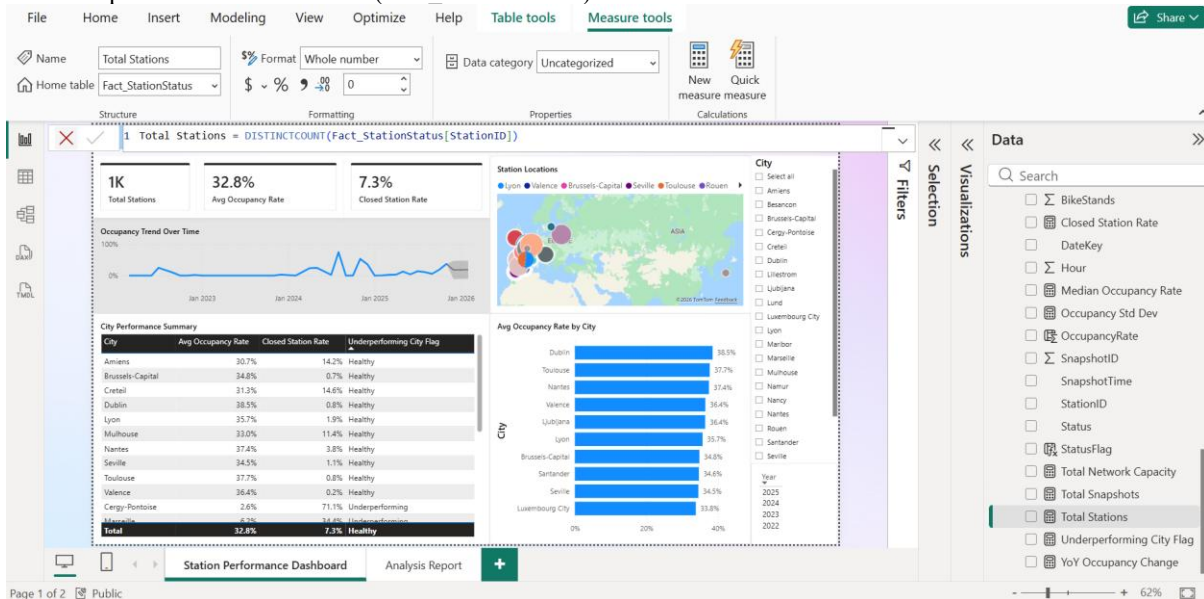
OccupancyRate = DIVIDE(Fact_StationStatus[AvailableBikes], Fact_StationStatus[BikeStands])



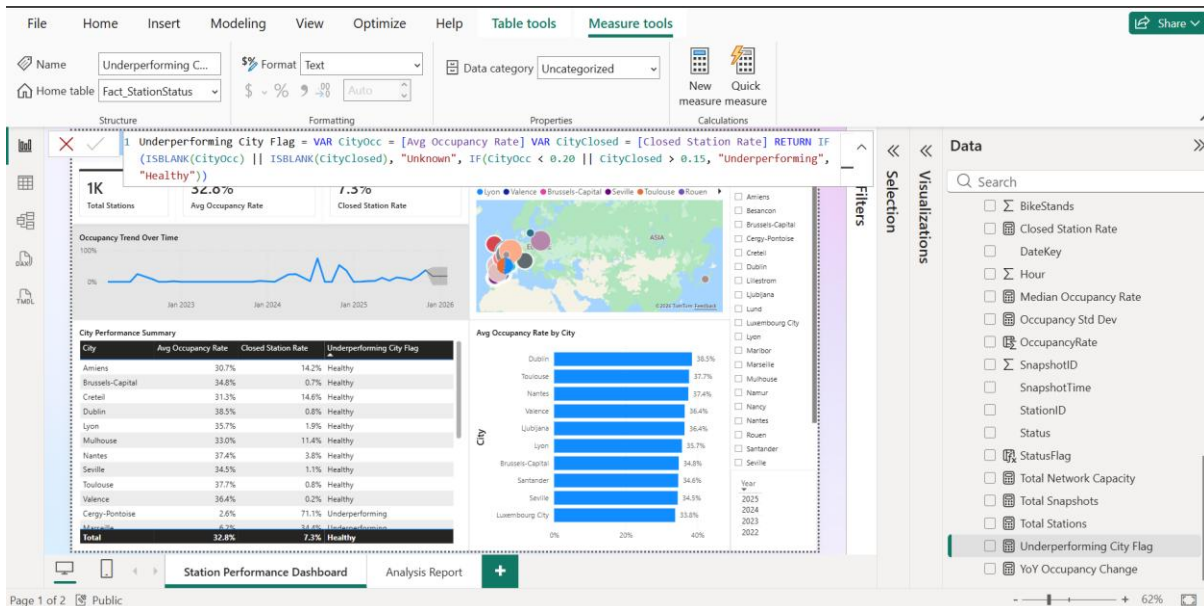
Total Network Capacity = SUMX(SUMMARIZE(Fact_StationStatus, Fact_StationStatus[StationID]), CALCULATE(AVERAGE(Fact_StationStatus[BikeStands])))



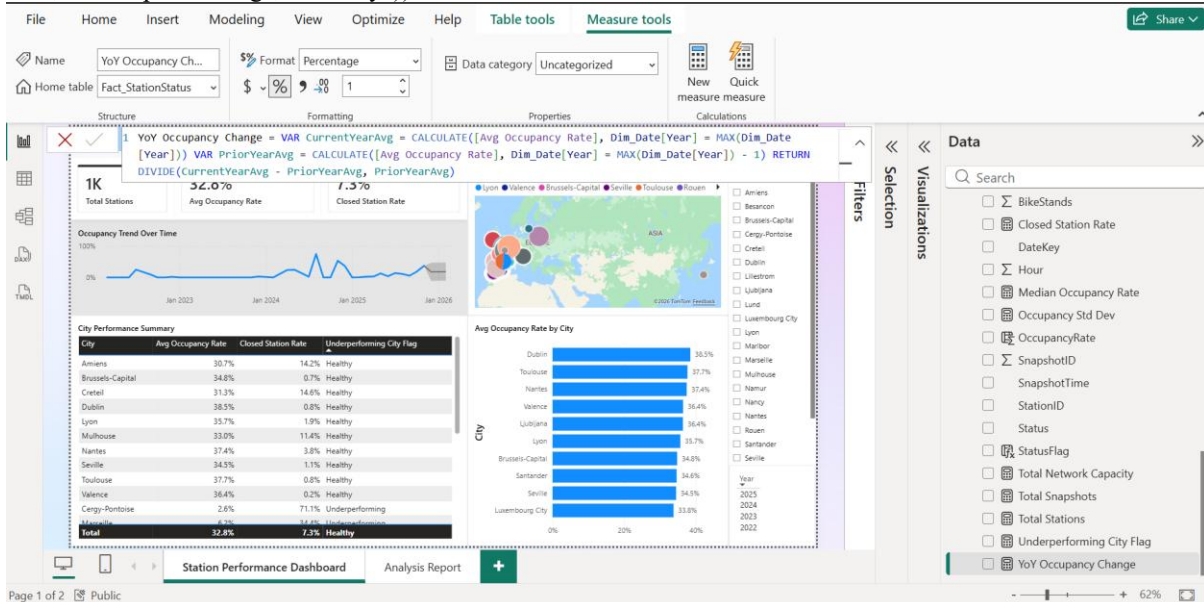
Total Snapshots = COUNTROWS(Fact_StationStatus)



Total Stations = DISTINCTCOUNT(Fact_StationStatus[StationID])



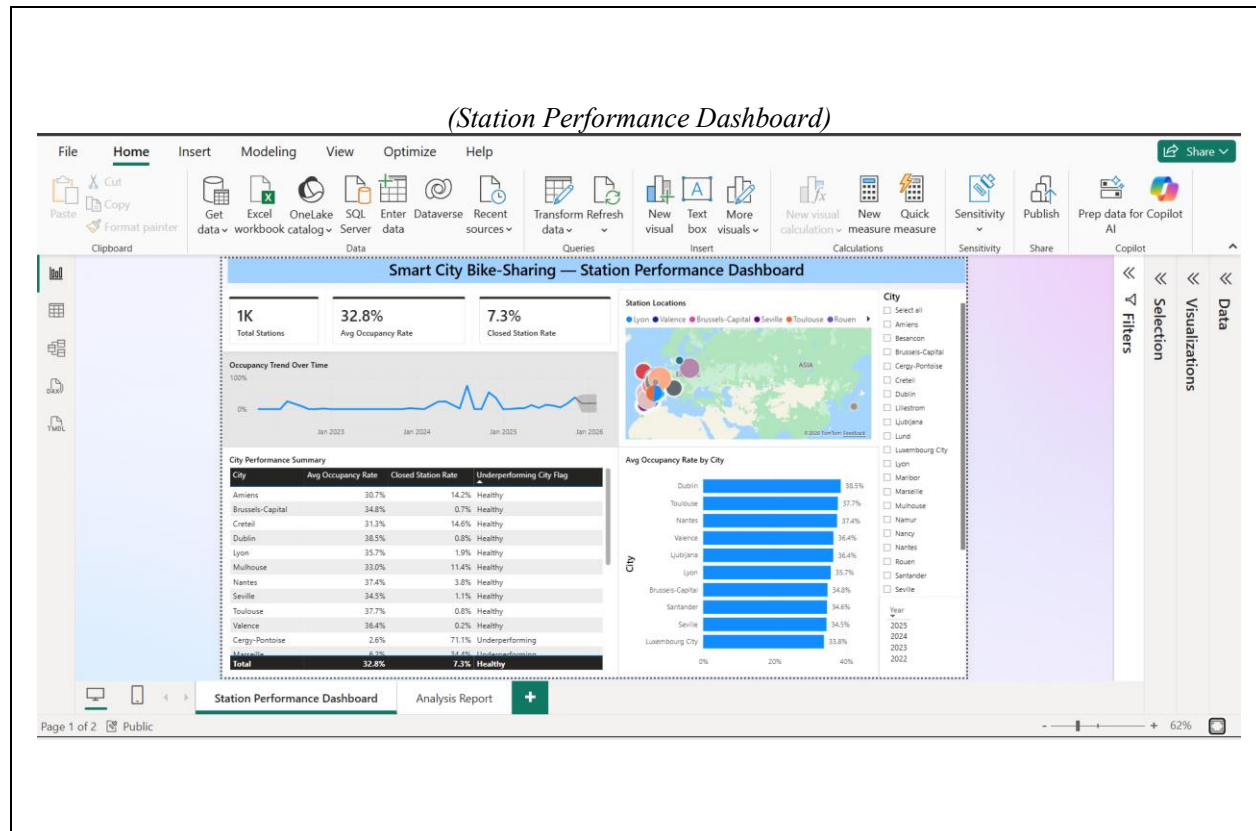
Underperforming City Flag = VAR CityOcc = [Avg Occupancy Rate] VAR CityClosed = [Closed Station Rate] RETURN IF (ISBLANK(CityOcc) || ISBLANK(CityClosed), "Unknown", IF(CityOcc < 0.20 || CityClosed > 0.15, "Underperforming", "Healthy"))



YoY Occupancy Change = VAR CurrentYearAvg = CALCULATE([Avg Occupancy Rate], Dim_Date[Year] = MAX(Dim_Date[Year])) VAR PriorYearAvg = CALCULATE([Avg Occupancy Rate], Dim_Date[Year] = MAX(Dim_Date[Year]) - 1) RETURN DIVIDE(CurrentYearAvg - PriorYearAvg, PriorYearAvg)

Visualization and Insights Arrived with Screenshot of Visualization / Report / Dashboard

Two report pages were built: a Station Performance Dashboard (KPI cards, station map, occupancy trend line with forecast, city performance table, bar chart, and City/Year slicers) and a four-quadrant Analysis Report page covering the four types of analysis below.



Descriptive Analysis - What is happening?

Focus: Summarizes historical data.

- Network-wide average occupancy rate is 32.8%, across 24 cities and all recorded snapshots.
- Dublin leads the network at 38.5% average occupancy; Cergy-Pontoise records the lowest.
- KPI cards summarize Total Stations, Avg Occupancy Rate, and Closed Station Rate at a glance.

Diagnostic Analysis - Why did it happen?

Focus: Identifies trends and patterns.

- Cergy-Pontoise's closure rate (71.1%) is far above the network average (7.3%), pointing to an operations/maintenance issue rather than low rider demand.
- Marseille (34.4% closed) shows a similar, though less severe, pattern.
- Namur and Rouen show moderate occupancy with elevated closure rates, suggesting partial rather than total station failure.

Predictive Analysis - What is likely to happen?

Focus: Uses data to forecast outcomes.

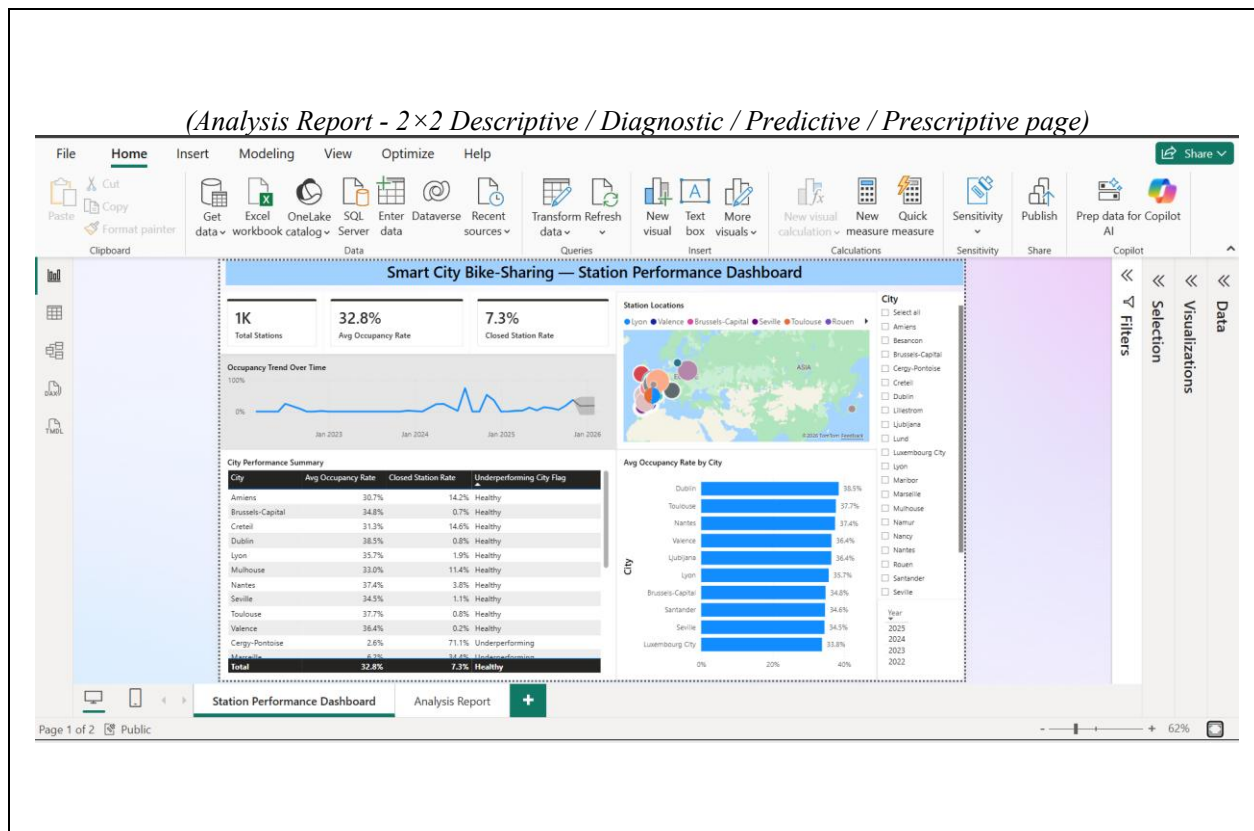
- A monthly occupancy trend line with a forecast band (built using Power BI's Analytics pane) projects near-term occupancy.

- Based on the 2022–2025 trend, occupancy is projected to continue fluctuating within the historical range for the next few months.

Prescriptive Analysis - What should we do about it?

Focus: Recommends actions to improve performance.

- Prioritize station audits and maintenance in Cergy-Pontoise and Marseille.
- Rebalance bike/dock allocation in Namur and Rouen.
- Benchmark uptime practices from Dublin, Toulouse, and Nantes across the rest of the network.
- Investigate cities flagged "Unknown" to confirm whether thin data reflects a small deployment or a data-collection gap.



Conclusion

This project demonstrates an end-to-end Power BI workflow - from importing and cleaning raw station-level data, through star-schema data modeling and DAX-driven measures, to an interactive, insight-oriented dashboard. The analysis shows a network operating at a healthy 32.8% average occupancy overall, but with clear operational gaps: Cergy-Pontoise and Marseille require immediate maintenance attention due to unusually high closure rates, while Dublin, Toulouse, and Nantes represent best-practice benchmarks the rest of the network can learn from. Acting on these recommendations - targeted audits, bike/dock rebalancing, and cross-city benchmarking - offers a clear path to improving overall network reliability and rider experience.